

Diego Arteaga

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Education

California State University, Northridge

Northridge, CA

Master of Science in Data Science

May 2027

Bachelor of Science in Computer Science (Summa Cum Laude)

May 2025

- **Relevant Coursework in:** Data Structures, Algorithm Design, Software Engineering, SQL Database Design, Big Data, Python for Data Science & Analytics, Data Visualization, Machine Learning, Deep Learning, Probability & Statistics

Experience

Outreach Representative

Sept 2025 - Present

Los Angeles Valley College

Los Angeles, CA

- Cleaned, maintained, and updated confidential LACCD student records using PeopleSoft and **Microsoft Excel**, performing data validation, recruitment reporting, and ensuring data integrity for the **LAVC Outreach database**
- Represented the college at high schools and community youth centers, assisting prospective students

Social Media Manager & Content Creator

Nov 2021 - Present

YouTube (Self-Employed)

Los Angeles, CA

- Analyzed content performance and managed YouTube Shorts, TikTok, and Instagram for a 300k+ subscriber creator brand (Arrowstotle), optimizing publishing strategy and audience engagement
- Built **Google Sheets** dashboards to track KPIs, monitor trends, and support data-driven content strategy decisions
- Independently managed personal pages and conducted **A/B tests** on content formats and thumbnail designs across monetized personal channels, increasing revenue by **22%** and engagement by **40%**

Undergraduate Research Intern

June 2025 - Aug 2025

California State University, Northridge

Northridge, CA

- Developed components of a deep learning-based fusion model for anomaly detection on healthcare IoT traffic data using **Python**, **PyTorch**, and **Scikit-learn** in **Google Colab**; preprocessed and analyzed large-scale IoT network data
- First author of an **IEEE-published research paper** and presented findings at the IEEE IEMCON 2025 conference

Projects

Healthcare IoT Network Anomaly Detection | *Python, PyTorch, Scikit-learn, Google Colab*

- Built an anomaly detection pipeline using Transformer-based architectures and temporal-graph feature engineering to identify malicious behavior in healthcare IoT networks
- Project recognition: **Best Paper Presentation, IEEE IEMCON 2025**

Japanese-to-English Fusion Translation Model | *Python, PyTorch, Pandas, Linux, Transformers, HuggingFace*

- Processed and analyzed a 3 million sentence bilingual corpus to evaluate Japanese-to-English translation performance using the SacreBLEU toolkit for BLEU, chrF, and BERTScore metrics
- Fine-tuned and evaluated language models including mBART, mT5, and OPUS-MT in a **Linux GPU environment**

Yoda RAG Model | *Python, Scikit-learn, NLTK, NLP, LLMs*

- Designed a Retrieval-Augmented Generation (RAG) chatbot using a curated Star Wars corpus, implementing TF-IDF vectorization and cosine similarity retrieval to generate contextually relevant Yoda-style responses
- Fine-tuned a pretrained language model using Low-Rank Adaptation (LoRA) to efficiently adapt response generation while reducing computational overhead

Agricultural IoT & Machine Learning Platform | *React, Bootstrap, Node.js, Python, Postman, Firebase*

- Designed a real-time dashboard using React and Bootstrap to visualize environmental sensor data from Firebase within a team of 12 students via GitHub and Jira, integrating and testing **Python backend APIs**
- Collaborated on machine learning models to correlate environmental data with plant sickness and crop growth
- Project recognition: **1st Place, CSUNposium 2025**

Skills

Programming Languages: Python, Java, JavaScript, SQL, HTML/CSS

Technologies: Git, React, Node.js, Express.js, NumPy, Pandas, Matplotlib, Scikit-learn, PyTorch, Jupyter Notebook, Google Colab, VS Code, MySQL, Google BigQuery, AWS, Jira, NPM

Languages: English, Spanish, Japanese